

Directions: Students MUST sign up (by Friday, April 10th) for an oral exam time between Tuesday, April 14th and Friday, April 17th. You will sign up for a 30-minute timeframe (though we will expect a 20-minute oral exam).

Students may collaborate with (only) their classmates to prepare for the oral exam, but only until Tuesday, April 14th at 11 AM, when the oral exams begin. Each student is allowed to bring in two pieces of 8.5" by 11" paper with notes (handwritten) to refer to, as well as a calculator and any appropriate software output. (Dr. Herman will provide the formula sheet from the in-class exam as well). (You may also ask Dr. Herman questions.) Canvas is the only website a student should access for help with the exam.

Students will be asked questions based on two of the following questions. Students will get to choose one of the problems she will base questions on while Dr. Herman will choose the other. Students may be doing a mix of writing (while explaining) and just explaining.

Problem 1: Generally, when you take kids down the cereal aisle, it appears that they are always drawn to the high sugar cereals (like Fruit Loops and Reese's Puffs). I decided to take random samples of the cereals on the shelves in the cereal aisle and record the sugar content (in grams) per serving as well as which shelf (bottom, middle, top). Use 10% significance level/90% confidence when deemed appropriate. (See **Cereals** data on Canvas)

- FULLY (all ways) assess the validity conditions of the ANOVA test to compare the average sugar content of the cereals on the different shelves. Be sure to provide all appropriate info and interpretations (as well as output). Use Minitab to get summary info (n, mean, standard deviation/variances for each shelf) (**SOFTWARE**).
- Regardless of what you said in part (a), construct the ANOVA table by hand. (Round all values to 2 decimal places) (You may use the summary output (mean, n, std.dev., variance). (Note – you do NOT need to do the "formal" hypothesis test.)
- Make a decision on the appropriate test and construct/interpret 90% simultaneous confidence intervals for the mean differences by-hand (using output you already have and this website (<https://statistics.tools/critical-value-t-calculator>) to help).
- There are many other ways to assess equal variances of the different populations. One such way is Levene's test.

This test performs an ANOVA on the absolute residuals, $|y_{ij} - \bar{y}_{i\bullet}|$. That is, the residual is the absolute value of the difference between an observation and the sample mean for that particular treatment. The Levene's test performs an ANOVA F-test using the absolute residuals in place of the original data. You will be performing this test by calculating the absolute residuals and running an ANOVA on them. I want to see the data used, a by-hand calculation of SST, and the output. Finally, compare this p-value to the one that you obtained in part (a) to test equal variances. (**SOFTWARE**).

Problem 2: An NFL kicker has historically made 48% of all field goals of 45 (or more) yards. During practices, they have been attempting 4 kicks for these distances at a time. We will presume that the number of kicks they make could be considered binomial using their historical accuracy. We are going to take a random sample of 100 instances (where they attempt 4 kicks) and record the number they make.

- Let p_i represent the probability that they make i of the 4 kicks. Determine what the null hypothesis will be for each of the values of p_i .
- From our random sample of 100 practices, we got the results below. Test to see if the data collected contradicts the Binomial distribution claimed with $\alpha = 0.10$.

Number of shots made (Y)	0	1	2	3	4
Frequency	10	16	42	22	10

- Would you want to construct any follow-up confidence intervals? If so, which and why (and then construct them); if not, explain why not.
- Find the overall sample proportion of kicks made (use the data). Construct and interpret a 90% confidence interval for the proportion of kicks made. Does this interval give you evidence that the kicker's rate of 48% has changed? Why or why not? Does this surprise you? Why or why not?

Problem 3: One may wonder if and how the relationship of your parents could affect your own future relationship. A study was done in 2024 on a sample of 40 – 50-year-olds in the United States. Each person in the sample was asked about how they would describe the relationship of their parents while they were growing up. We classified this into two groups: Married/Together or Not Together. In addition, each person was asked about their relationship status, giving them four possible options: single (never married), married, divorced, or widowed. The results are summarized in the table below:

	Individual's Relationship Status			
Parents' Status	Single	Married	Divorced	Widowed
Married/Together	80	259	130	31
Not Together	84	94	105	17

- Is there evidence of a relationship between the two variables? Use a 1% significance level, showing all steps.
- Based on the results above, which, if any, relationship statuses of the individuals do you believe there are big differences in when it comes to parents' status? Explain.
- Calculate and interpret a 99% confidence interval to compare the proportion of those that are married for those whose parents are married/together to those whose parents are not together.
- Calculate and interpret a 99% confidence interval to compare the proportions of those who are married to those who are single for those whose parents are still together.

Problem 4: I believe that those who shop for weekly groceries inside Kroger will spend more money than those who schedule delivery. To test this, I am going to sample 4 people who scheduled delivery and 7 people who shopped inside Kroger and recorded the amount of money that they spent.

- We want to determine a rejection region based on W through direct calculation. I want my significance level to be no more than 3%. For what values of W should we reject H_0 , assuming there are no ties?
- Now write out the hypotheses and determine the rejection region using the appropriate table.
- Verify/show that these rejection regions (a) and (b) are equivalent!

Problem 5: For ANOVA, prove that $SS = SST + SSE$. (Be able to EXPLAIN this process, not just "do" it!)